UNIVERSITÀ DI TORINO

CRYSTAL

Modelling Diffusion Coefficients and Ionic Transport

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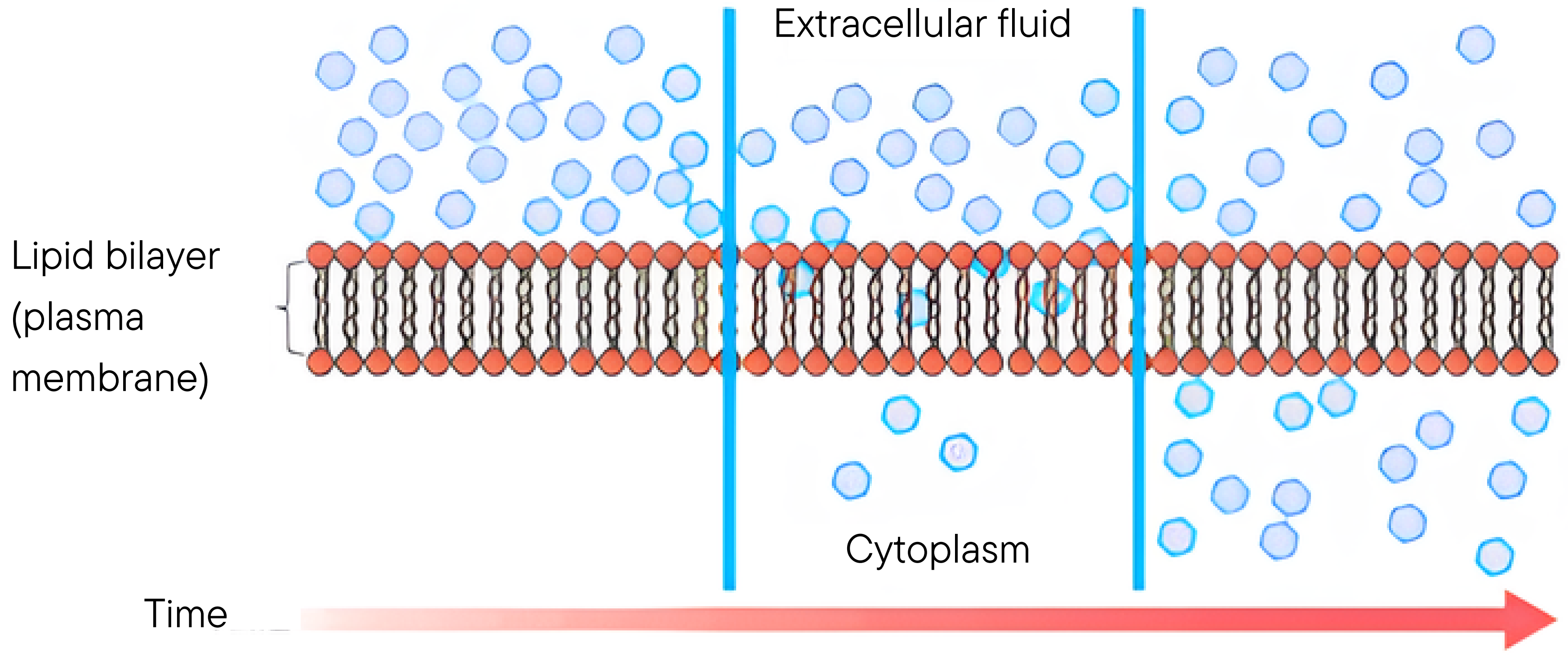
Table of Contents

Diffusion Coefficient: What Is It?

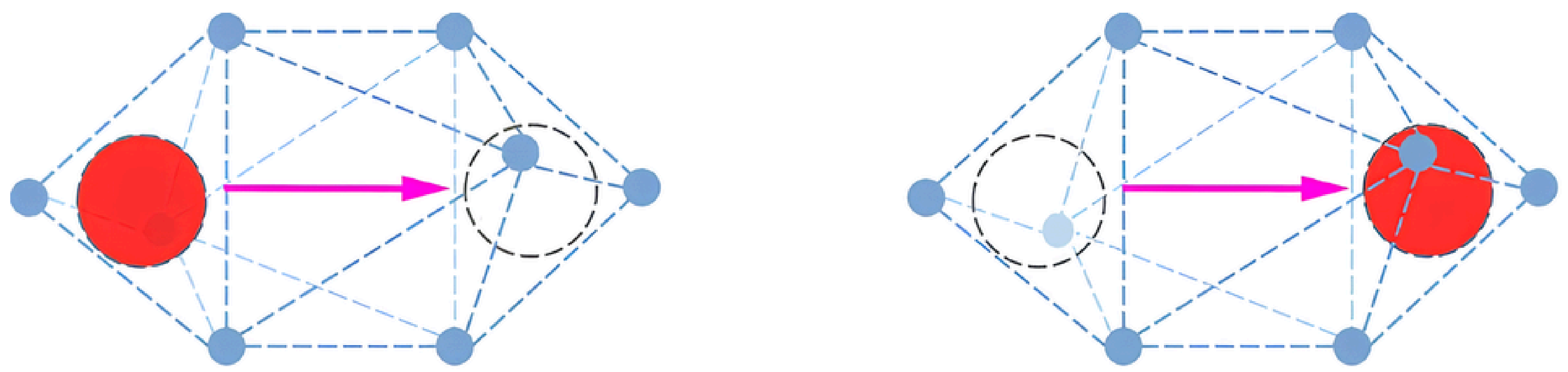
The **diffusion coefficient** represents the **amount of a particular substance** diffusing across a **unit area** through a unit concentration gradient in a **unit time** → $\mathbf{m^2\,s^{-1}}$ or $\mathbf{cm^2\,s^{-1}}$

Transport in a liquid



https://www.labxchange.org/library/items/lb:LabXchange:96ccd65a:html:1

Transport in a solid



He, X. et al., Nat Commun 8, 15893 (2017)

Diffusion Coefficient: What Is It?

Diffusion of atoms and ions is an important kinetic property of many materials → it determines whether stable states can be reached and at which rate this can occur

Low diffusion rates

Prevent evolution to an equilibrium state (e.g. to avoid corrosion of a material)

High diffusion rates

Fast ionic diffusion desirable (e.g. solid electrolytes and insertion electrodes)

Van der Ven, A. et al., Phys. Rev. B, 64, 184307

Diffusion Coefficient: What Is It?

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Ionic diffusion in crystalline solids typically occurs by **diffusion-mediating defects** such as **vacancies** or **interstitial sites**

Dilute regime ← **Low carriers concentration**

High carriers concentration → **Non-dilute regime**

Van der Ven, A. et al., Phys. Rev. B, 64, 184307

Diffusion Coefficient: How Is It Calculated?

There is **not** a **standard** way to calculate the diffusion coefficient

- **Nudged Elastic Band (NEB)** → find the minimum energy diffusion paths

  Henkelman, G.; et al. J. Chem. Phys. 2000, 113 (22), 9901-9904

- **Molecular dynamics** → solve Newton's equations of motion to generate the trajectory for a system over a time period at finite temperature and pressure

Time scales and sizes

Classical
- **ns - µs**
- **thousands atoms**

Ab initio
- **ps**
- **hundreds atoms**

- **Kinetic Monte Carlo** → numerical calculation of the diffusion coefficient by stochastic simulations of the migration of a collection of ions within a host

Van der Ven, A. et al., Phys. Rev. B, 64, 184307

Diffusion Coefficient: How Is It Calculated?

Dilute regime

Dilute diffusion theory

$$D \propto a^2 \nu^* exp\left(-\frac{\Delta E_a}{k_B T}\right)$$

- a $\longrightarrow$ hop distance
- ΔE_a $\longrightarrow$ activation barrier
- ν^* $\longrightarrow$ hopping frequency
- T $\longrightarrow$ temperature
- k $\longrightarrow$ Boltzmann constant

Gladstone, S.; et al., The Theory of Rate Processes; McGraw-Hill, 1941

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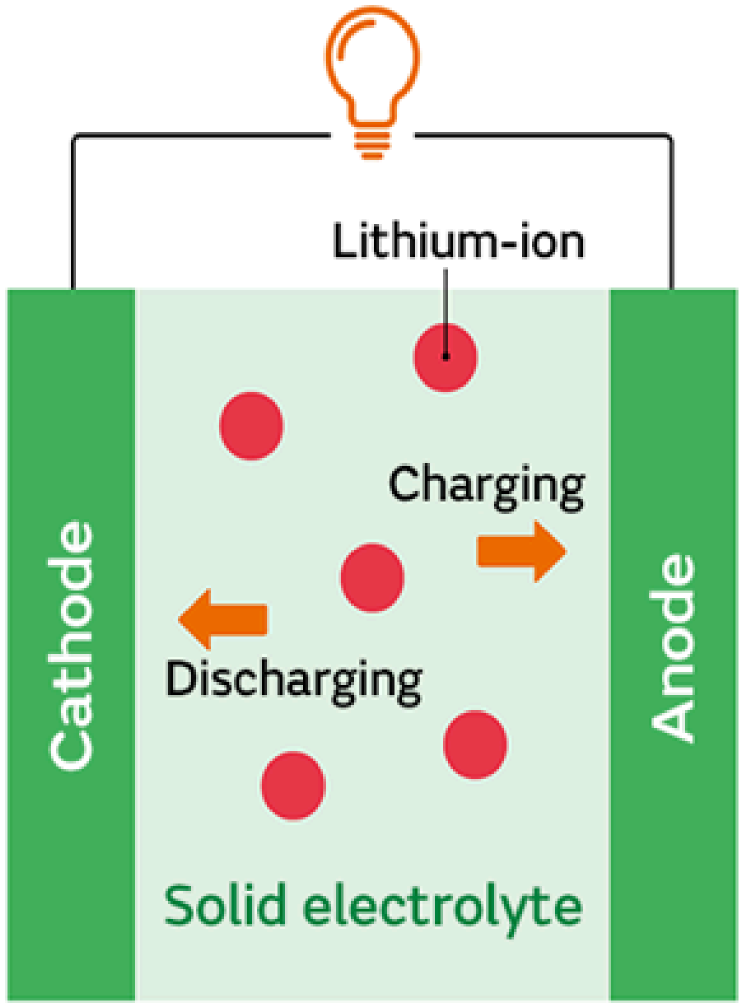
Separation in different components that can be individually modelled

→ Deeper insight into diffusion process

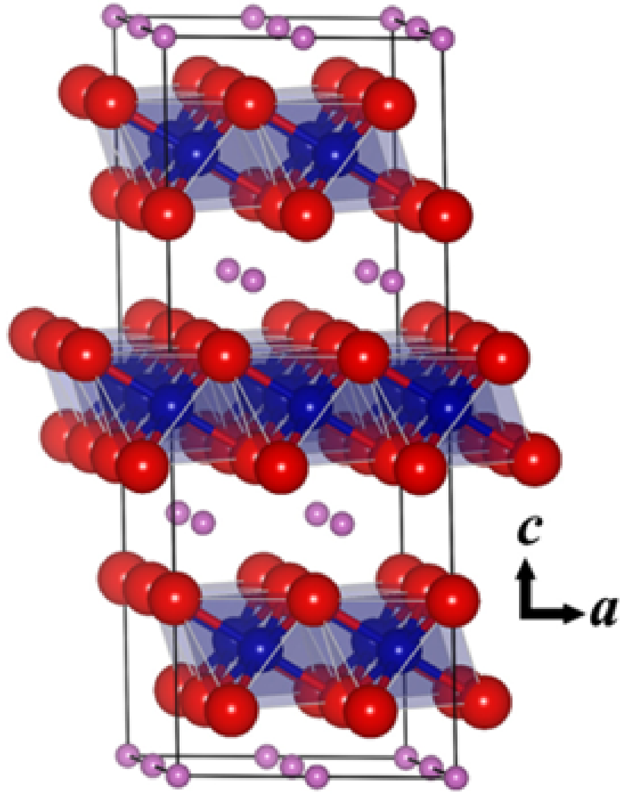
Gladstone, S.; et al., The Theory of Rate Processes; McGraw-Hill, 1941

Case Study

Modelling Li-ion diffusion coefficient in cathode materials for Li-ion batteries



Transition metal layered oxides



Li diffusion crucially affects the performance of the battery

Case Study

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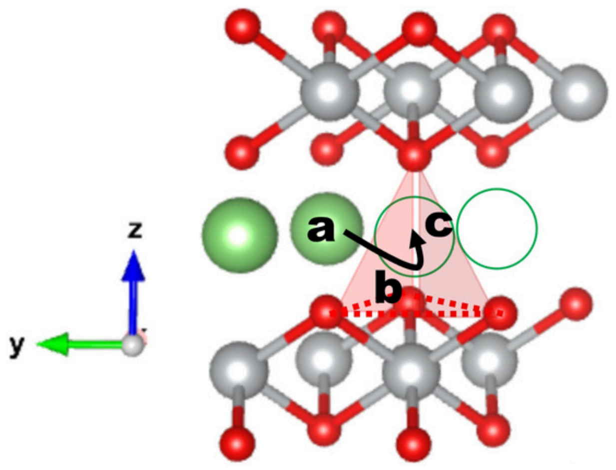
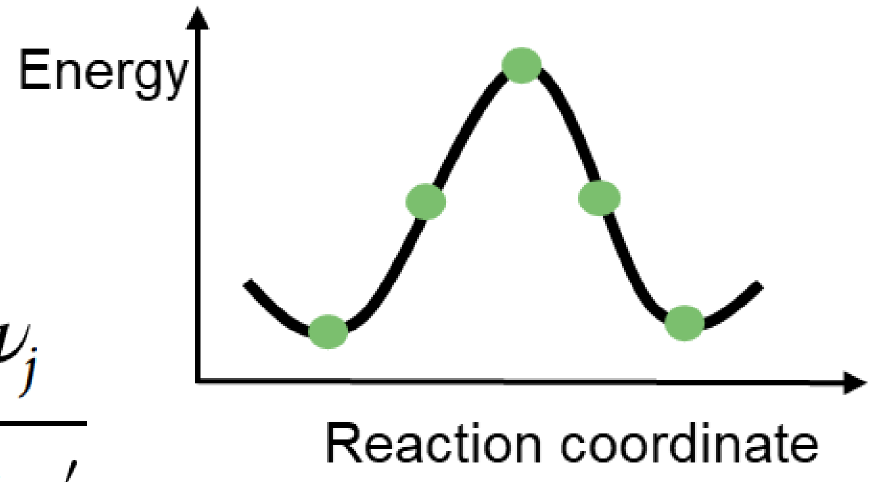


$$\nu^* = \frac{\prod_j^N \nu_j}{\prod_j^{N-1} \nu_j'}$$

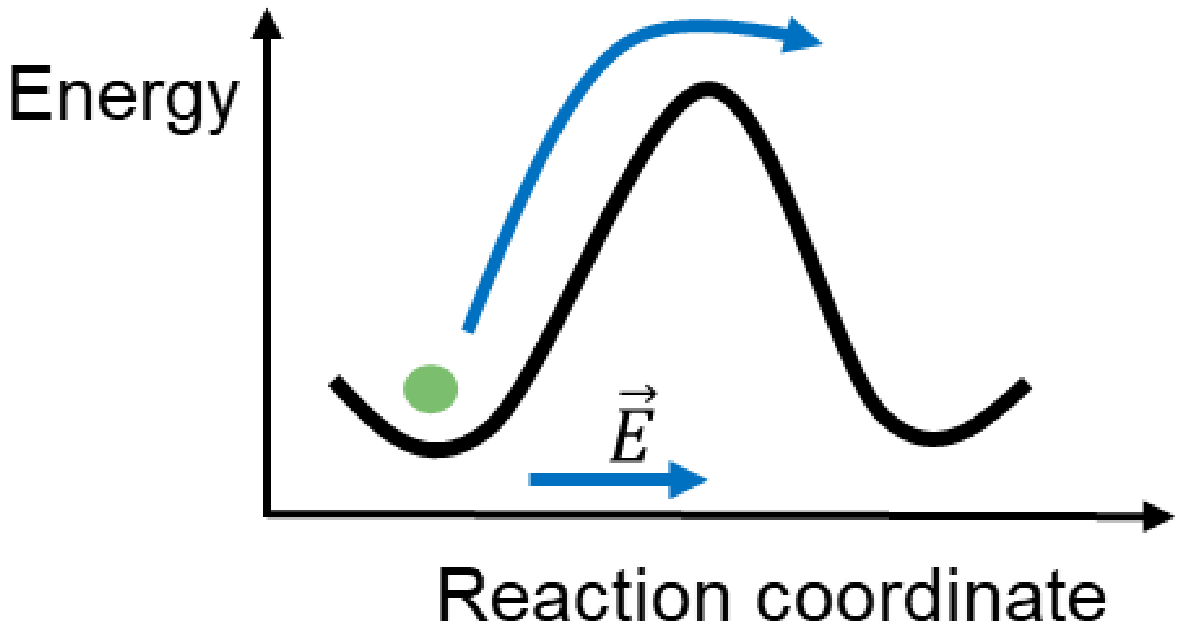


There is **not** a **standard** way to calculate the diffusion coefficient

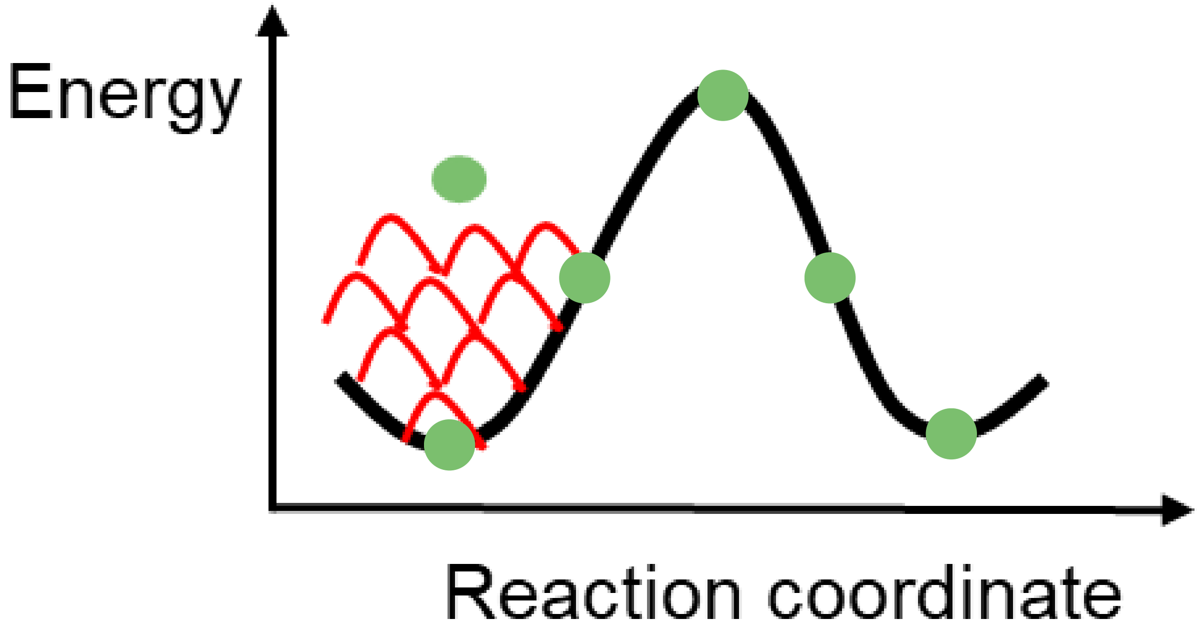
Ion Path and Activation Barrier Calculation: Methods

$$D \propto a^2 \nu^* exp\left(-\frac{\Delta E_a}{k_B T}\right)$$

Ab initio molecular dynamics



Metadynamics + path-sampling

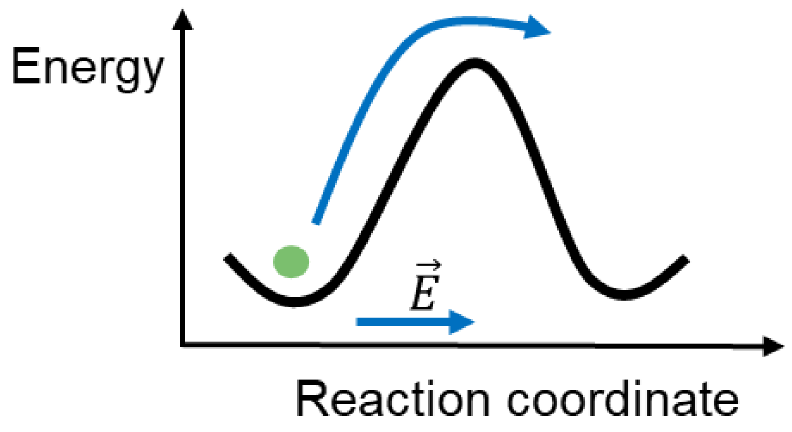


Bonometti, L. et al, J. Phys. Chem. C 2024, 128, 11979-11988

Ion Path and Activation Barrier Calculation: Methods

Ab initio molecular dynamics

Ion motion
- **Temperature increase**
- **Application of an external potential**



- Constant temperature
- Canonical ensemble
- ps time scale
- Potential applied along a specific direction
- Evaluation of mean-square displacement as a function of the potential

Bonometti, L. et al, J. Phys. Chem. C 2024, 128, 11979-11988

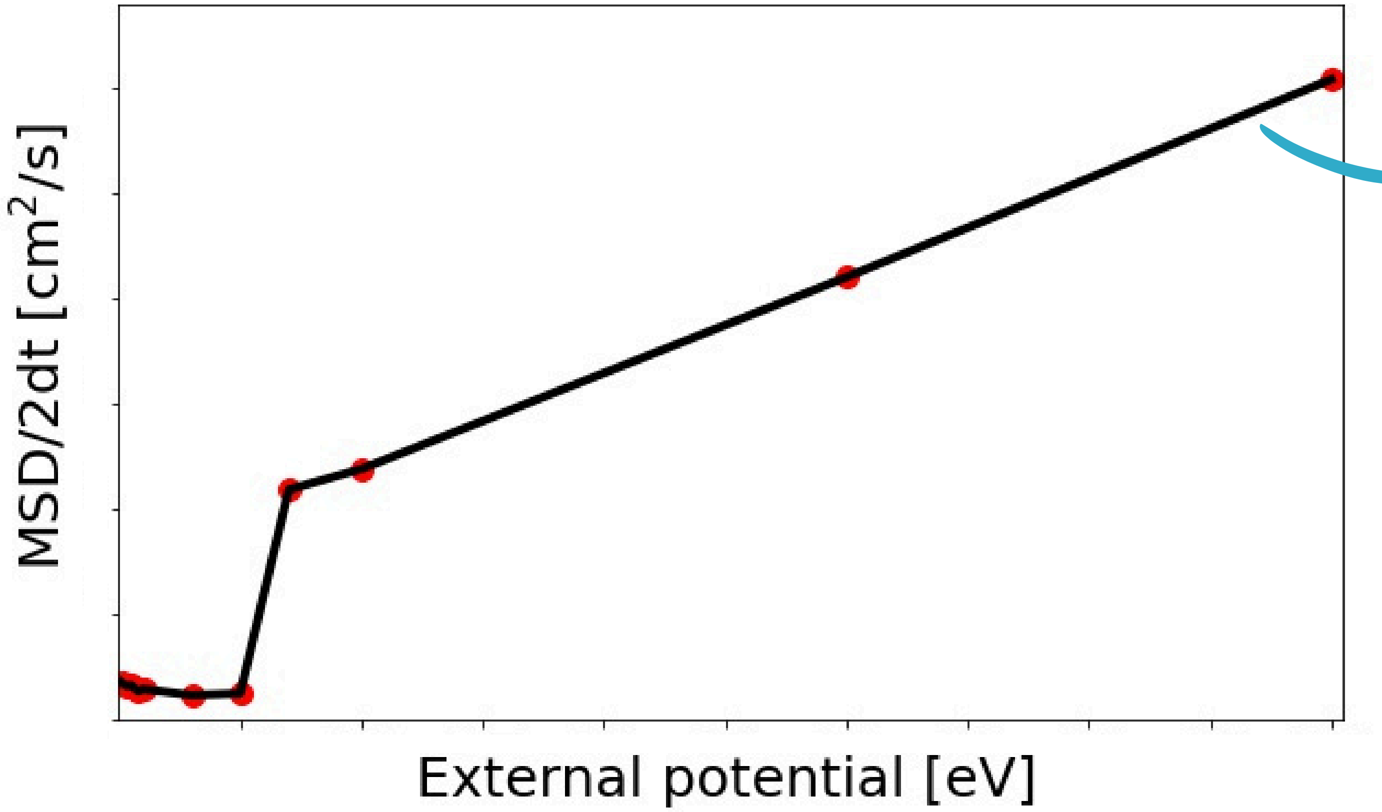
Ion Path and Activation Barrier Calculation: Methods

Ab initio molecular dynamics

Einstein's relation

$$\frac{1}{2d}\lim_{\tau\to\infty}\frac{\langle \|p(t)-p(t+\tau)\|^2\rangle_t}{\tau}$$

Einstein, A. Ann. Phys., 1906, 324, 371-381



Extraction of activation energy from this trend and ion path from the simulated trajectory

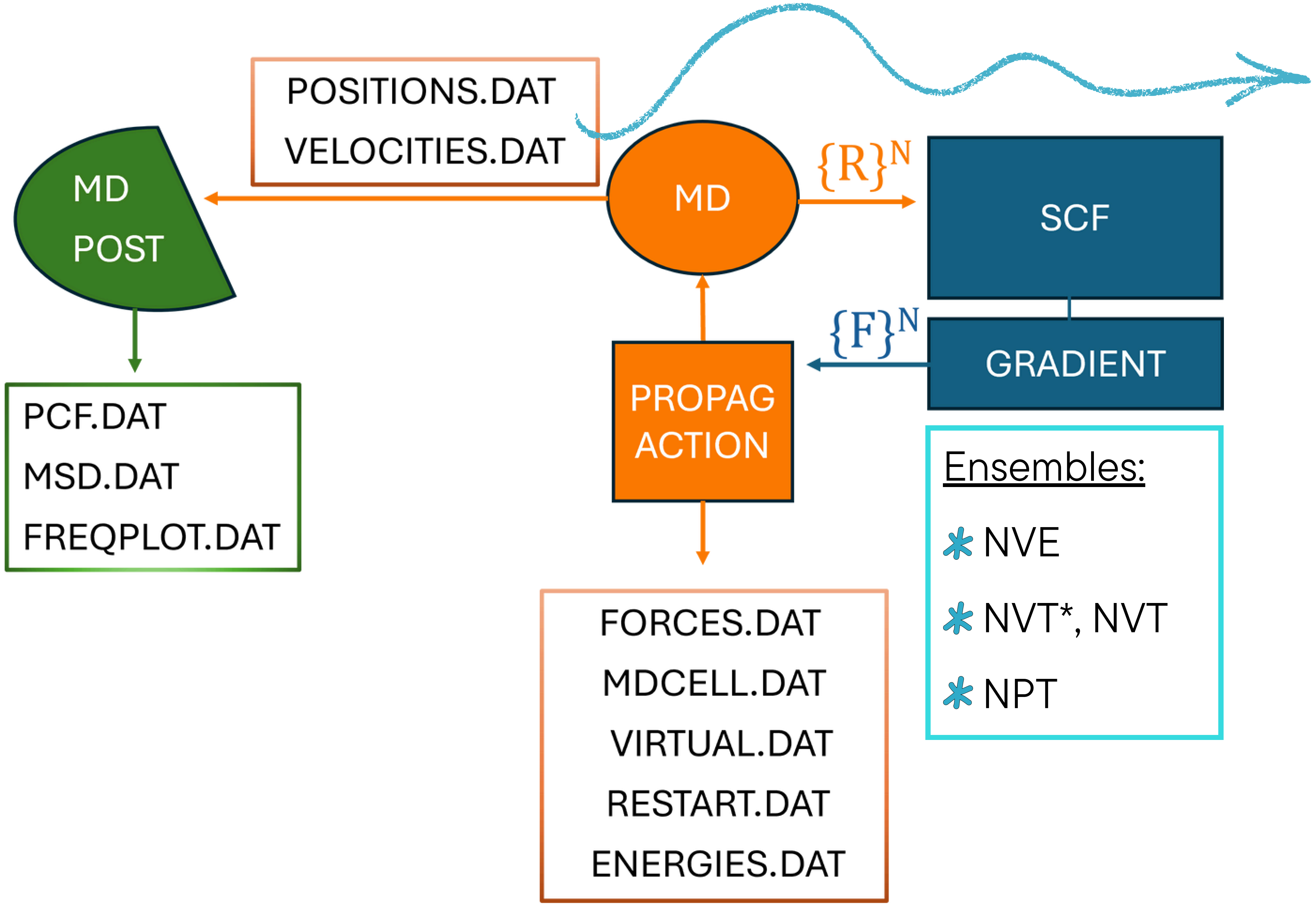
key ingredients for D

Calculations performed before the implementation of molecular dynamics module in CRYSTAL, so another program was exploited

Bonometti, L. et al, J. Phys. Chem. C 2024, 128, 11979-11988

NEW Molecular dynamics module in CRYSTAL

Calculation of diffusion coefficient from post-processing treatment of molecular dynamics data



* From positions:

$$D = \frac{1}{2d} \frac{\partial \left\langle R^2(t) \right\rangle}{\partial t}$$

* From velocities:

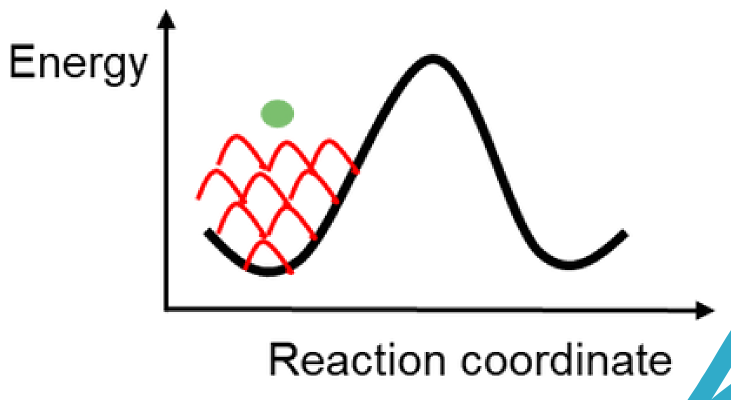
$$D = \int_0^\infty \langle v(t) v(t+\tau) \rangle d\tau$$

Recent applications:

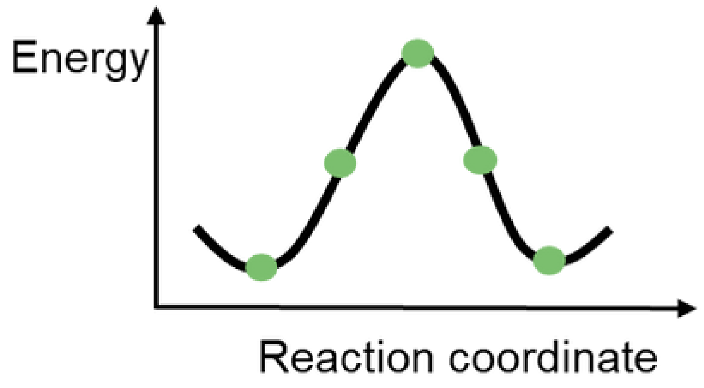
self-diffusion coefficient of Li and ionic conductivity (NVT esemble) for Li_3PS_4 and Li_2S

Ion Path and Activation Barrier Calculation: Methods

Metadynamics + path-sampling



- Sampling method by the addition of an external history-dependent bias potential
- **Reconstruction of the path** with less constraints than other approaches
- Energy barries values overestimated because of the intrisic limitations of the method



- Sampling of the path determined via metadynamics
- Calculation of single-point energies
- Reconstruction of energy curve and **calculation of energy barrier** with higher accuracy

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Ion Path and Activation Barrier Calculation: Methods

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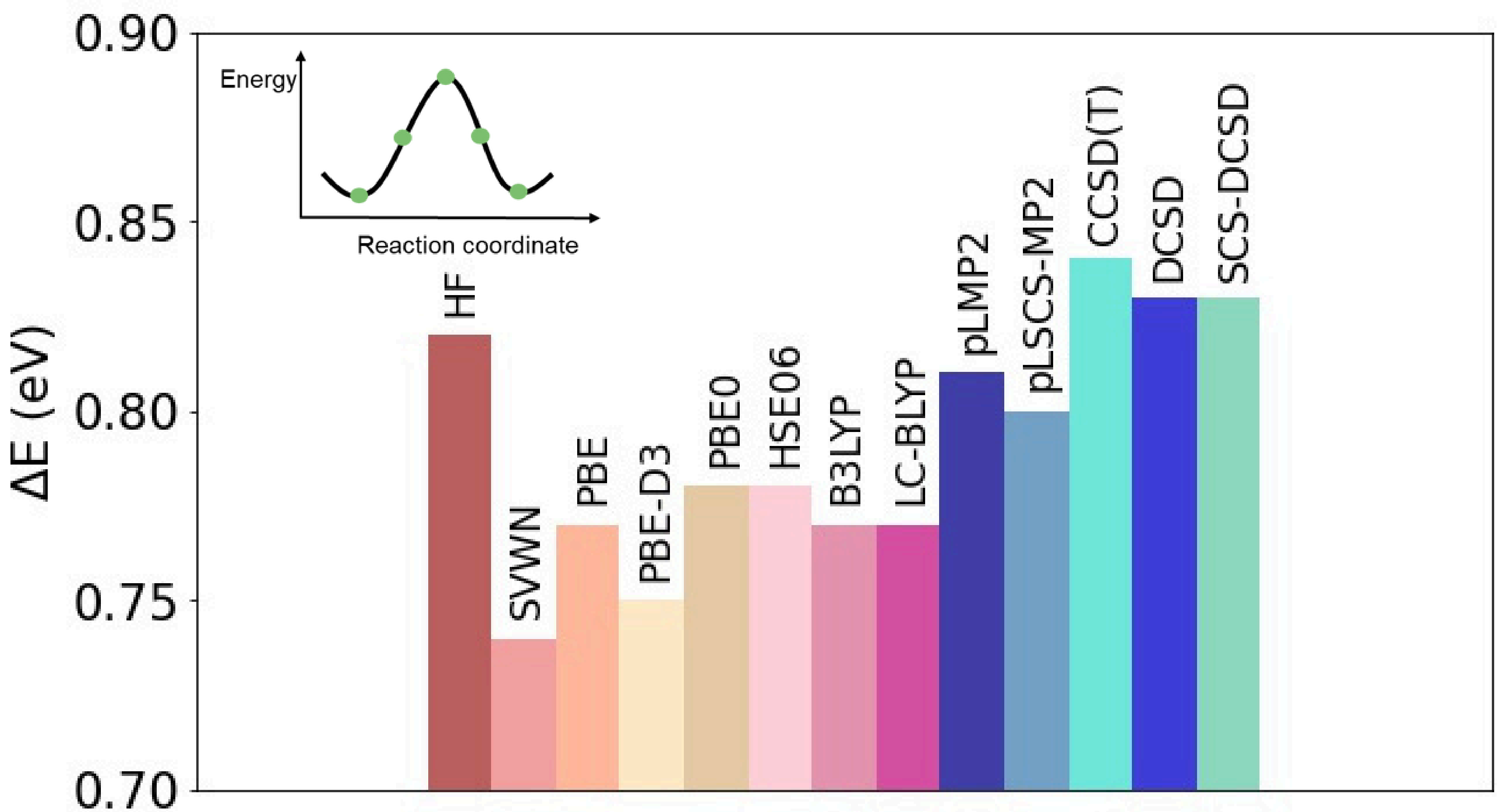

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CRYSCOR A program for electron correlation in crystals

- **Weak sensitivity** of barrier height to the use of different DFT functionals/post-HF methods in **single-point energy differences**
- Mechanism governing the diffusion energetics mainly dominated by **electrostatics**
- Van der Waals dispersion and electron correlation effects to a large extent cancel out

Bonometti, L. et al, J. Chem. Theory Comput. 2024, 20, 10114-10119

Hopping Frequency Calculation

$$D \propto a^2 \nu^* exp\left(-\frac{\Delta E_a}{k_B T}\right)$$

$$\nu^* = \frac{\prod_j^N \nu_j}{\prod_j^{N-1} \nu'_j}$$

Vineyard, G. H. J. Phys. Chem. Solids 1957, 3, 121-127.

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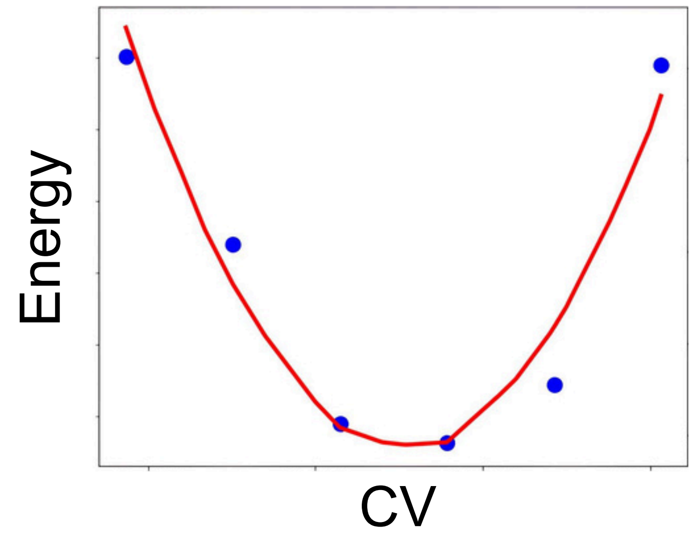
- Calculation of vibrational frequencies in equilibrium state
- Optimization of transition state geometry
- Calculation of vibrational frequencies in transition state

- Fit the minimum corresponding to the initial equilibrium state with a parabola
- 1-dimensional harmonic oscillator approximation
- One frequency for the ground state, and no real frequencies left at the maximum



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Summary

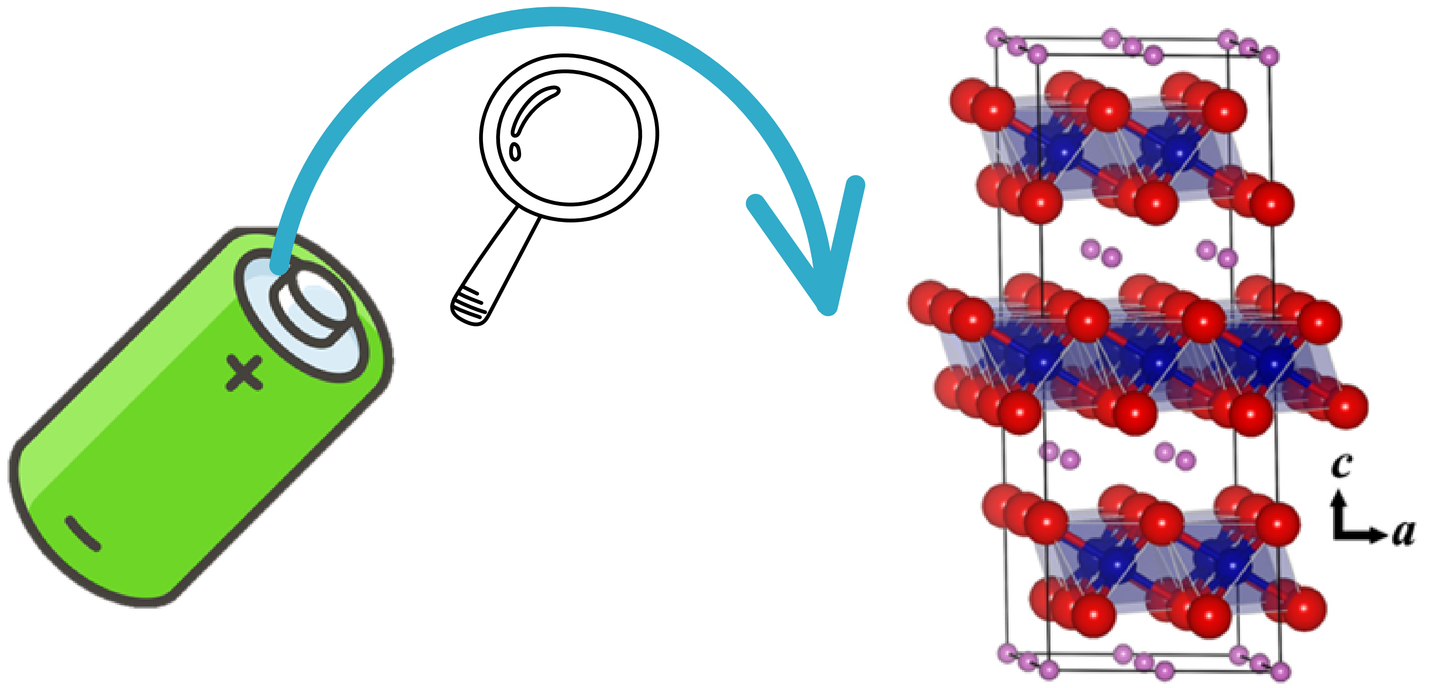
* Diffusion coefficient → key property of many technologically relevant materials

* **Dilute diffusion theory**

$$D \propto a^2 \nu^* exp\left(-\frac{\Delta E_a}{k_B T}\right)$$

- Hop distance
- Hopping frequency
- Activation energy

How can we model D?



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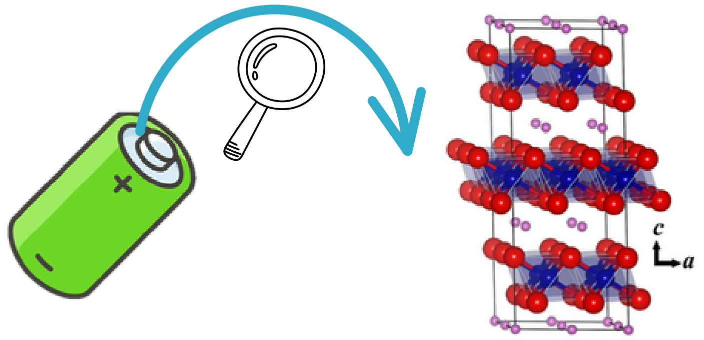
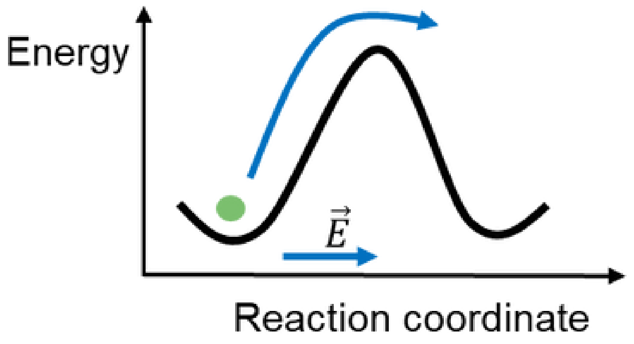
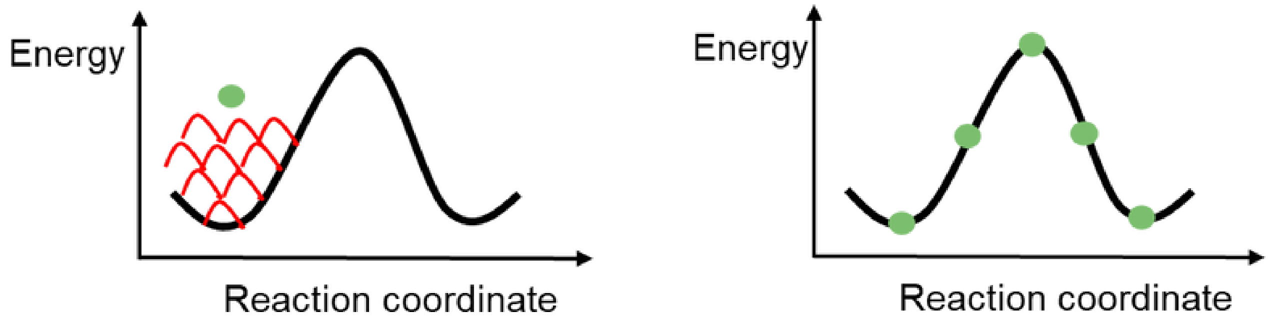


***Ab initio* molecular dynamics** CRYSTAL



Metadynamics + path-sampling CRYSTAL



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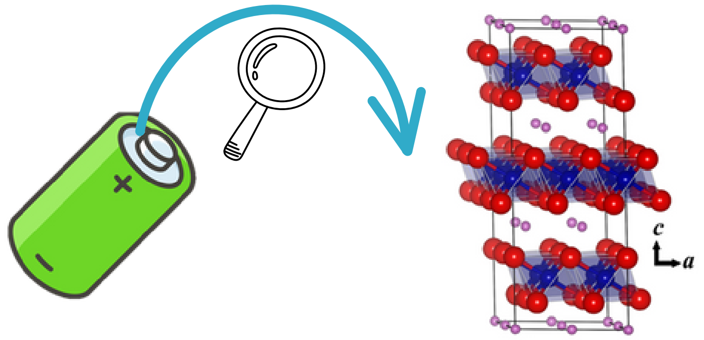
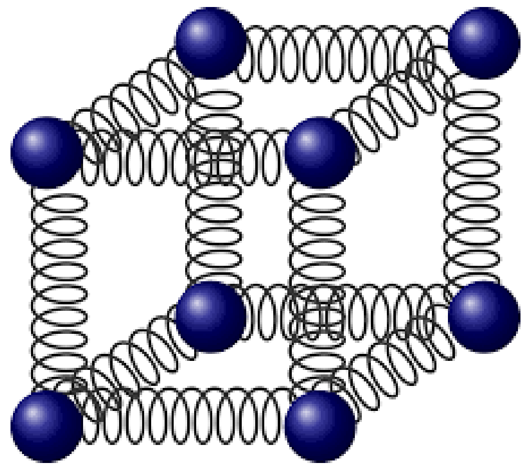
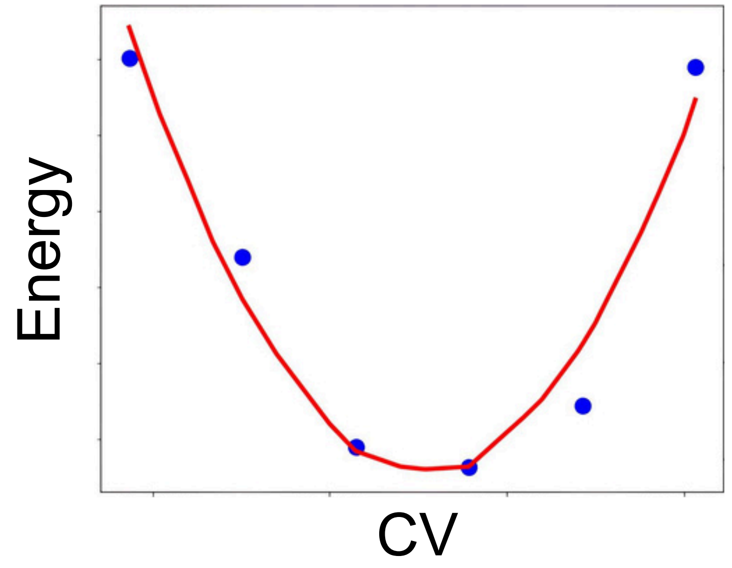


Frequencies calculation

CRYSTAL



1D approximation



Thank you for your attention!